

Workforce Analytics Project

A CRISP-DM Analysis of Attrition, Compensation and Performance

Prepared by	Rosemary Waweru
Role	People Analytics Specialist
Deliverable	Project Summary Document (CRISP-DM Report)
Companion file	TalentCore_Workforce_Analytics.pbix — 5-page Power BI dashboard
Dataset	TalentCore HR Workforce Data (1,000 raw rows → 243 analysed employees)
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Executive Summary

TalentCore employs 243 people across six departments and six East African offices. This report analyses the company's workforce data to explain why people leave, whether pay is equitable, and how performance varies across teams. The analysis follows the CRISP-DM methodology and is supported by a five-page interactive Power BI dashboard.

Three findings drive the recommendations that follow. First, attrition tripled in 2024, rising from 2.1% to 5.1% in a single year. Second, the employees most likely to leave are not the overworked but the disengaged — staff who work no overtime leave at three times the rate of those who do. Third, performance is weaker than headline attrition suggests: more employees are rated 1–2 than 4–5, and the company average is 2.89 out of 5.

Headline metrics

Metric	Value	Metric	Value
Total headcount	243	Average monthly salary	KES 150,807
Active employees	205 (84.4%)	Median monthly salary	KES 141,000
Employees who left	38	Gender pay gap	0.9%
Overall attrition rate	15.6%	Average performance rating	2.89 / 5
Average tenure	4.8 years	Employees rated 1–2	40.3%
Average tenure at exit	3.5 years	Estimated cost of attrition	KES 34.4 million

Estimated attrition cost assumes a replacement cost of six months' salary per departure, a widely used industry benchmark. See Section 6 for the assumptions underpinning each figure.

1. Business Understanding

1.1 Organisational context

TalentCore is a pan-African technology services company with offices in Nairobi, Mombasa, Kisumu, Kampala, Dar es Salaam and Kigali. Its workforce spans six departments: Engineering, Sales, Finance, Human Resources, Operations and Customer Support. The company has grown steadily since 2018, from 40 employees to 217 at the end of 2025.

Leadership has committed to data-driven people decisions but currently has no single view of workforce health. Attrition, compensation and performance data sit in a single operational extract that has never been analysed systematically.

1.2 Business questions

The analysis was scoped around six questions posed by the People function:

1. How many people does TalentCore employ, and how has headcount changed over time?
2. What is the attrition rate, and is it rising or falling?
3. Which departments, offices and employee segments carry the highest attrition risk?
4. Is compensation equitable across gender, department and education level?
5. How does performance vary across teams, and does training investment influence it?
6. What actions should leadership take to improve retention in the next 12 months?

1.3 Stakeholders and success criteria

Stakeholder	Primary interest	What success looks like
Chief HR Officer	Company-wide retention and cost	A single executive view with quantified, actionable recommendations
Department heads	Team-level performance and risk	Ability to filter to their own department and see their own numbers
Finance	Payroll cost and headcount planning	Reliable compensation totals to support the 2026 budget
People Operations	Data quality and reporting	A repeatable, documented pipeline rather than manual spreadsheet work

The project is considered successful if every business question above can be answered from the dashboard without further analysis, and if each answer is traceable to a documented transformation.

2. Data Understanding

2.1 Dataset overview

The source is a single Excel worksheet, TalentCore HR Workforce Data.xlsx, containing 1,000 rows and 16 columns. The data dictionary supplied with the brief describes 15 fields; an undocumented sixteenth column, Remarks, was also present.

Field	Type	Description	Completeness
Employee ID	Text	Unique identifier	100% (4 duplicates)
Full Name	Text	Employee name	100%
Hire Date	Date	Date joined	100%
Exit Date	Mixed	Date left; blank if active	14.2% populated
Department	Text	Business unit	100%
Job Role	Text	Role within department	100%
Office Location	Text	City	100%
Gender	Text	Gender	100%
Age	Mixed	Age in years	95.9%
Education Level	Text	Highest qualification	100%
Monthly Salary (KES)	Mixed	Gross monthly salary	96.3%
Performance Rating	Mixed	Most recent rating, 1–5	97.9%
Training Hours	Numeric	Hours in last 12 months	91.4%
Overtime	Mixed	Works overtime regularly	100%
Attrition Status	Text	Active or Left	100%
Remarks	Text	Free-text notes (undocumented)	6.6%

2.2 Data quality audit

Initial profiling revealed that the dataset is substantially unclean. The single most consequential finding is that 753 of the 1,000 rows contain a Hire Date and nothing else — every other field is blank. These are padding rows, not employees. Loading the file without removing them would have inflated headcount fourfold and computed every average over a majority of nulls.

Field	Issues identified	Examples
Employee ID	Leading/trailing spaces, inconsistent case, stray decimal suffix, 4 duplicate IDs	" EMP-1100 ", emp-1164, EMP-1285.0
Exit Date	Six date formats, five text placeholders, Excel serial numbers stored as text	2024.11.02, 13-Dec-2024, 03 May 2023, 05/27/2025, 44804, "Not Applicable", "Still Active", "-"
Department	Case variants, abbreviations	Tech, Ops, Support, Cust. Support
Office Location	Case variants, abbreviations, misspellings	DSM, Niarobi, Mombassa
Gender	Four encodings	Male, male, MALE, M

Field	Issues identified	Examples
Age	Text suffix, impossible values	"57 yrs", -5, 200
Monthly Salary	Currency prefix, thousands separators, one negative value	"KES 242,800", "97,700", -105,900
Performance Rating	Three encodings	4, 4/5, "4 out of 5"
Training Hours	Three negative values	-56, -41, -17
Overtime	Six encodings	Yes, yes, Y, TRUE, 1 / No, no, N, FALSE, 0
Attrition Status	Case variants and synonyms	Currently Employed, Resigned, Terminated
Education Level	Fifteen variants of four levels	Bachelors, BSc, "bachelor's degree", MSc, Masters, DIPLOMA, Certification

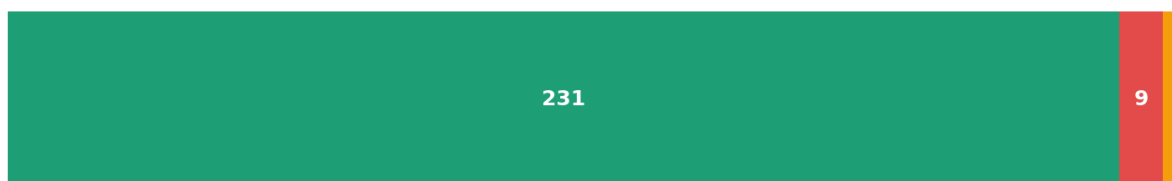
2.3 Logical contradictions

Beyond formatting, the data contained internal inconsistencies that no amount of type conversion would resolve:

- Nine employees have an Exit Date earlier than their Hire Date, one as early as October 2017 — before the company's first recorded hire.
- Three employees are marked as having Left but carry no Exit Date, making their tenure and exit timing unmeasurable.
- Four Employee IDs appear twice, each pair identical except for the Hire Date.
- Four employees fall outside any plausible working age, recorded as -5 or 200.

Encouragingly, one contradiction that was tested for did not occur: no employee marked Active carries an Exit Date. Testing for a problem and finding none is itself a result worth recording.

■ Clean records (231)
 ■ Exit before hire (9)
 ■ Missing exit date (3)



243 records retained · 95.1% clean · 753 padding rows and 4 duplicate IDs removed before analysis

Figure 1 — Composition of the analysed dataset after cleaning

3. Data Preparation

All transformations were performed in Power Query so that the pipeline is repeatable and auditable. No values were edited manually in the source file.

3.1 Transformation log

#	Transformation	Rationale	Effect
1	Filtered out rows with a null Employee ID	Removes padding rows carrying only a Hire Date	1,000 → 247 rows
2	Trimmed and cleaned all text columns	Removes leading/trailing whitespace and control characters	All text fields
3	Standardised Employee ID to uppercase, removed ".0" suffix	Enables reliable duplicate detection	247 rows
4	Mapped category variants to canonical values	Six departments, six offices, two genders, four education levels	Department, Office Location, Gender, Education Level, Overtime, Attrition Status
5	Stripped currency symbols and separators from Salary	Converts text to a numeric type	234 valid values
6	Extracted the leading digit from Performance Rating	Normalises "4/5" and "4 out of 5" to 4	238 valid values
7	Parsed Exit Date across six formats and five placeholders	Converts a mixed-type column to a true date	35 dates recovered
8	Removed duplicate Employee IDs, retaining the earliest Hire Date	Conservative tenure treatment; documented rule	247 → 243 rows
9	Nullled out-of-range values (Age, Salary, Training Hours)	Removes impossible values without discarding the employee	14 values nullled (10 age and 8 salary values later imputed — see step 11)
10	Nullled Exit Dates earlier than Hire Date	Removes logically impossible dates	9 values nullled
11	Imputed missing Age and Salary with the column mean	Keeps all 243 records in banded and average analysis rather than isolating them in an "Unknown" category	10 ages, 8 salaries filled
12	Added Data Quality Flag column	Retains a record of which rows had issues	12 records flagged
13	Added Salary Band, Age Band and sort-order columns	Enables banded analysis with correct chart ordering	5 bands each

3.2 Documented assumptions

Several cleaning decisions required judgement. Each is recorded here so that a reviewer can disagree with the choice rather than be misled by it.

Assumption	Reasoning	Alternative considered
Slash-format dates are MM/DD/YYYY	One value is 05/27/2025; there is no month 27, so the format must be US-order	DD/MM/YYYY — rejected as impossible

Assumption	Reasoning	Alternative considered
Duplicate IDs resolved by keeping the earliest Hire Date	Treats the record as the original hire and yields the conservative tenure figure	Keeping the most recent — would overstate churn
Missing Age and Salary imputed with the column mean	Mean imputation preserves the company average exactly while allowing all 243 records to appear in banded analysis. Leaving them null placed 8 employees in an “Unknown” salary band and 10 in an “Unknown” age band, distorting both distributions	Leaving values null — rejected because the “Unknown” category was read as a real segment
Negative Training Hours nulled, not converted to absolute values	A sign-entry typo cannot be confirmed; inventing values is worse than losing three	Number.Abs() — rejected
Exit dates before hire dates nulled; employee retained	The employment record is valid even where the exit date is not	Deleting the row — would lose four valid employees
Training hours of zero treated as real, not missing	Zero training is a meaningful business fact and a candidate attrition driver	Nulling zeros — would hide six employees

3.3 Date dimension

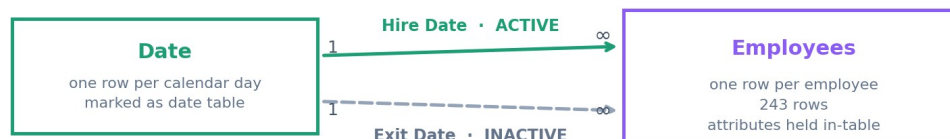
A calendar table was generated in DAX spanning the earliest date in the data through the current year, with Year, Quarter, Month, Month Year and sort-order columns. It was formally marked as a date table so that time intelligence resolves correctly. The calendar start date is derived from the earlier of the first hire and the first exit — necessary because the earliest exit date in the raw data precedes the earliest hire date.

4. Modelling

4.1 Schema design

The model is a two-table star: a single Employees fact table joined to a Date dimension. Department, Office Location, Gender and Education Level are held as attribute columns on the Employees table rather than being split into separate dimension tables. With 243 rows and low-cardinality attributes, normalising them would add joins without improving performance or clarity — a flat star is the appropriate design at this scale. Raw key columns are hidden from the report view and all measures are grouped in a dedicated measures table.

The design decision that most affects the analysis is the treatment of Date as a role-playing dimension. The Employees table contains two dates — Hire Date and Exit Date — but Power BI permits only one active relationship between any two tables. Hire Date holds the active relationship; Exit Date is joined by a second, inactive relationship that is activated inside individual measures.



Solid = filters flow automatically · Dashed = activated inside a measure using USERELATIONSHIP

Figure 2 — Star schema with a role-playing date dimension

From	To	Cardinality	Cross-filter	State
Date[Date]	Employees[Hire Date]	One to many	Single	Active
Date[Date]	Employees[Exit Date]	One to many	Single	Inactive

Without the second relationship, exits could not be placed on a time axis at all — a date slicer would have no path to reach the Exit Date column, and no attrition trend could be produced.

4.2 Key DAX measures

Thirty-nine measures were developed. The table below covers those that carry the analysis; the remainder are formatting and comparison variants.

Measure	Definition	Result
Total Headcount	DISTINCTCOUNT of Employee ID	243
Active Headcount	Headcount filtered to Attrition Status = Active	205
Employees Left	Headcount filtered to Attrition Status = Left	38
Attrition Rate	Employees Left ÷ Total Headcount	15.6%
Exits	Row count via the inactive Exit Date relationship, excluding blanks	26
Headcount as at Date	Hired on or before the date, and either still employed or exited after it	217 at end-2025
Annual Attrition Rate	Exits ÷ Headcount as at Date	5.1% in 2024

Measure	Definition	Result
Avg Tenure (Years)	Average of a calculated tenure column using exit date or today	4.8 years
Avg Monthly Salary	Average of cleaned salary	KES 150,807
Gender Pay Gap %	$(\text{Male average} - \text{Female average}) \div \text{Male average}$	0.9%
% High Performers	Share rated 4 or 5	32.9%
% Low Performers	Share rated 1 or 2, excluding blanks	40.3%
Data Completeness %	$\text{Records without a quality flag} \div \text{total records}$	95.1%

Two measures required particular care. Exits must explicitly exclude blank exit dates, because activating a relationship does not itself filter anything — without the exclusion, an unfiltered card returns all 243 rows. And % Low Performers must exclude blanks, because in DAX a blank value evaluates as less than or equal to 2, which would silently classify five unrated employees as low performers.

5. Evaluation

5.1 Validation approach

Every measure was validated against an independent recalculation of the source file before being placed on a dashboard page. Three errors were caught this way: an attrition trend that was routing through the Hire Date relationship rather than the Exit Date one, a low-performer measure that was counting blanks, and an exits figure that still included nine logically impossible dates.

5.2 Limitations

Limitation	Impact on the analysis
Only 243 usable records	Departmental and role-level rates rest on small denominators. Job-role attrition figures are based on 6–14 employees each and should be read as signals, not conclusions.
12 of 38 leavers lack a usable exit date	Time-based attrition charts plot 26 exits, not 38. Every trend figure therefore understates true attrition.
No exits recorded before 2021	Three years of apparently perfect retention is more likely an artefact of how the data was generated than a real result.
Snapshot, not longitudinal, data	Salary, performance rating and training hours are current values only. Year-on-year comparison of pay or performance is impossible.
2025 is a partial year	Hires run to May and exits to June 2025, so the final data point on every trend understates the full year.
No manager, engagement or exit-reason data	The analysis can identify where attrition occurs but not directly why. Recommendations therefore propose investigation where causation cannot be established.

5.3 Confidence assessment

Finding	Confidence	Basis
2024 attrition spike	High	Based on dated exits; the pattern is unambiguous
Overtime–attrition relationship	High	Large sub-samples (133 and 110 employees)
Tenure cliff at 1–3 years	High	65 employees in the affected band
Departmental attrition ranking	Medium	35–54 employees per department
Gender attrition gap	Medium	Balanced samples, but no controlling variables available
Job-role hotspots	Low	6–14 employees per role

6. Deployment

The deliverable is a Power BI Desktop report file containing the cleaned dataset, the data model and the five dashboard pages described in Section 7. Because the source is a static extract rather than a live system, this section records what has been delivered and what the analysis shows must change upstream, rather than proposing a hosting arrangement that the project has not tested.

6.1 What has been delivered

Component	Description
Power BI report file	Five dashboard pages plus a hidden employee drill-through page, with synchronised slicers and a reset control
Repeatable transformation pipeline	Twelve documented Power Query steps that re-run against any refreshed extract of the same structure, with no manual editing of the source file
Measure library	Thirty-nine DAX measures with defined business meanings, grouped in a dedicated measures table
Data quality monitor	A flag column and completeness measure surfaced on the executive page, so the condition of each new extract is visible rather than assumed

6.2 Required upstream changes

The analysis identified specific defects in how the data is captured. These are the highest-value deployment actions because they determine the quality of every future refresh.

Defect found	Required change
Exit Date stored in six different formats plus five text placeholders	Enforce a single date type on the field and reject free-text entries such as "Not Applicable" and "Still Active"
Nine exit dates precede the employee's hire date	Add a validation rule preventing an exit date earlier than the hire date
Three leavers have no exit date recorded	Make the field mandatory when Attrition Status is set to Left
Fifteen spellings of four education levels; similar variance across five other fields	Replace free-text entry with controlled dropdown lists
Four duplicated Employee IDs	Enforce uniqueness on the identifier
No exit reason captured for any of the 38 departures	Add a mandatory exit-reason field — without it, no analysis can move from where attrition occurs to why

Implementing the first three alone would eliminate nine of the twelve records currently flagged and raise data completeness from 95.1% to above 99%.

6.3 Maintaining the report

The pipeline re-runs end to end against a new extract without modification, provided the column names and structure are unchanged. Two conditions should be checked at every refresh: that total headcount matches the expected employee count rather than the raw row count, which would indicate padding rows have reappeared; and that the data quality panel has not deteriorated. Metric definitions should be reviewed whenever a new field is added to the source, since several measures depend on specific text values such as "Active" and "Left".

7. Dashboard Design

The report comprises five pages plus a hidden drill-through page. A consistent filter rail runs down the left of every page with slicers for date range, department, office location and attrition status, synchronised across all pages, together with a reset button. Each page is colour-coded so that a reader always knows which lens they are looking through.

Page	Question it answers	Accent	Principal visuals
1. Workforce Overview	Who do we have, and where?	Teal	KPI cards, headcount trend, department donut, hires vs exits, office and age distributions

Page	Question it answers	Accent	Principal visuals
2. Departmental Performance	Which teams are delivering, and at what cost?	Blue	Performance matrix with data bars, training vs performance scatter, rating by department
3. Attrition Analysis	Why do people leave?	Red	Attrition trend, rate by department, overtime comparison, rate by rating and tenure band
4. Compensation & Diversity	Is pay equitable?	Purple	Salary by department and gender, salary bands, tenure vs salary scatter, pay by education
5. Executive Summary	What should leadership do?	Mixed	KPI row, trends, insights and recommendations panel, data quality monitor

A drill-through page allows a reader to right-click any department and see the individual employees behind the number, with headcount, attrition rate and average salary recalculated for that department alone. This turns the attrition page from a static chart into something a department head can act on directly.

8. Key Insights

Insight 1 — Headcount has grown steadily, but 2024 broke the pattern

TalentCore grew from 40 employees at the end of 2018 to 217 at the end of 2025, adding between 29 and 40 hires a year with remarkable consistency. Exits, however, were not consistent. Having stayed at four or fewer through 2023, departures jumped to eleven in 2024 — nearly tripling in a single year while hiring stayed flat.

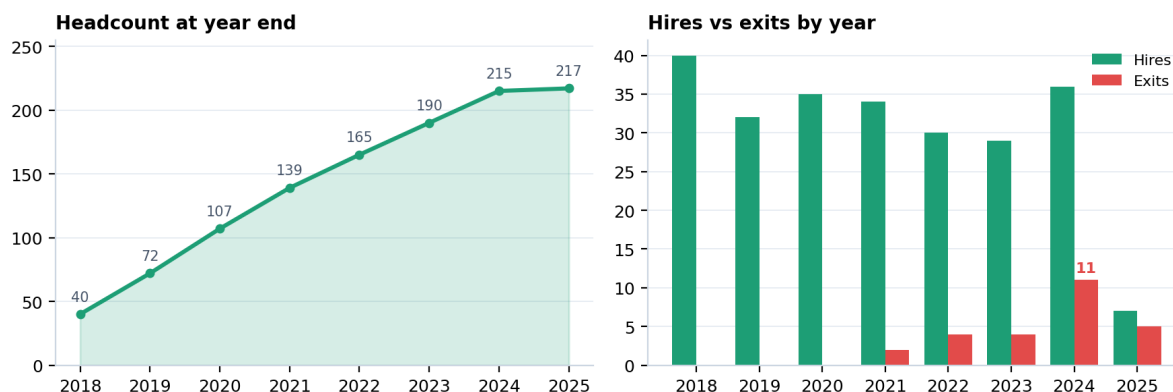


Figure 3 — Headcount growth and the hiring/exit balance, 2018-2025

Insight 2 — Attrition more than doubled in 2024

Expressed as a rate, annual attrition rose from 2.1% in 2023 to 5.1% in 2024. The company had been operating at a very low rate by industry standards; 2024 represents a break in that trend rather than a continuation of it. The 2025 figure of 2.3% appears to show recovery, but the data covers only half that year and should not be read as evidence of improvement.

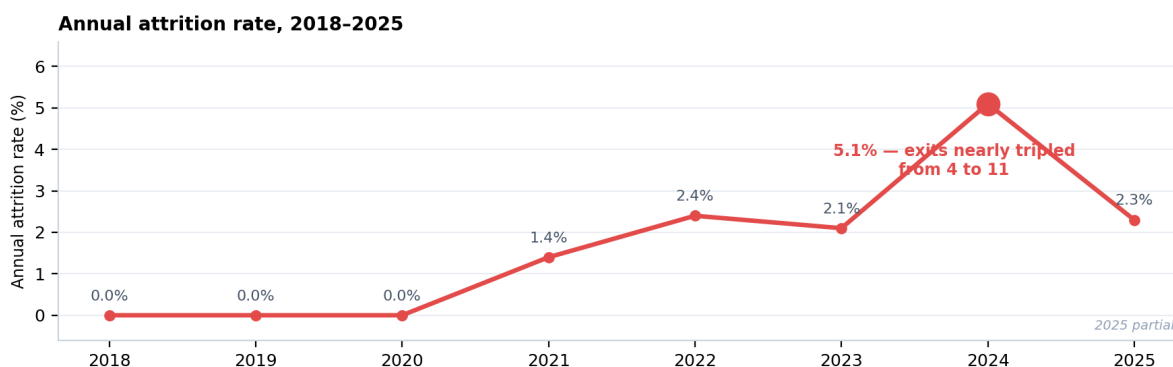


Figure 4 — Annual attrition rate, showing the 2024 break in trend

Insight 3 — Finance and Customer Support carry the risk; Nairobi does not

Attrition is not evenly distributed. Finance loses 21.6% of its people and Customer Support 20.4%, against just 8.1% in Operations. The pattern is not explained by pay: Engineering pays almost twice what Customer Support pays and still loses 17.1%, while Operations pays mid-range and retains best.

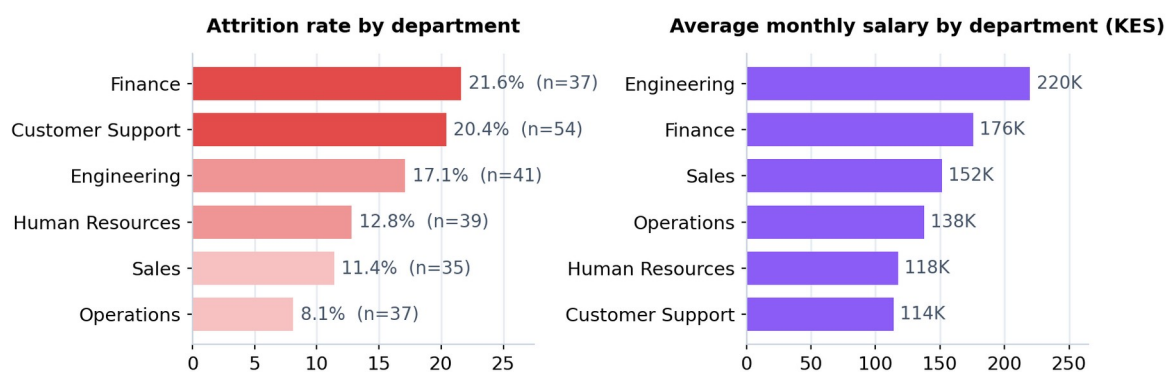


Figure 5 — Attrition rate and average salary by department

The geographic pattern is starker still. Nairobi loses 5.6% of its staff; Kisumu loses 22.9% — a fourfold difference between offices of almost identical size, with average salaries within KES 7,000 of each other. Whatever accounts for the difference, it is not compensation.

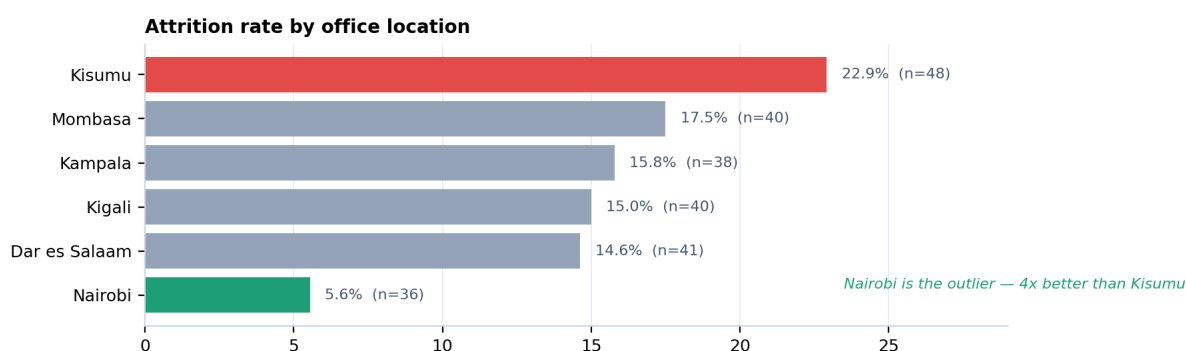


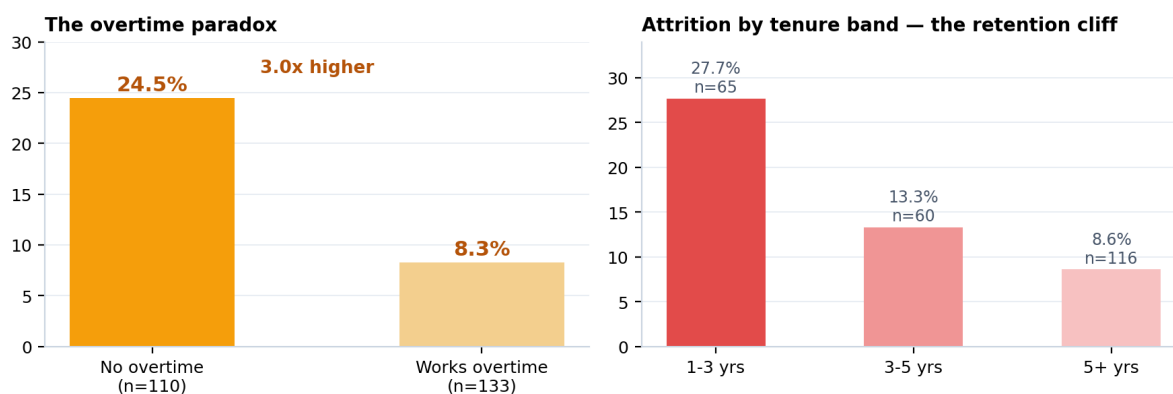
Figure 6 — Attrition rate by office location

Insight 4 — The overtime paradox, and a cliff at two years

The most counterintuitive finding in the dataset concerns overtime. Conventional wisdom holds that overworked employees burn out and leave. TalentCore's data shows the opposite: employees who work no overtime leave at 24.5%, three times the 8.3% rate among those who do. Both groups are large — 110 and 133 employees respectively — so this is not a small-sample artefact.

The most plausible reading is that overtime here is a proxy for engagement rather than for workload. Employees who are invested in their work take on additional hours and stay; those who are disengaged do neither. On this reading, low overtime is an early-warning signal rather than a sign of healthy work-life balance.

Tenure shows an equally clear pattern. Attrition runs at 27.7% among employees with one to three years of service, falling to 13.3% at three to five years and 8.6% beyond five. The company loses people precisely at the point where recruitment and onboarding investment has been made but not yet recovered.



Insight 5 — Performance is the deeper problem

Attrition is the presenting symptom, but the performance distribution is arguably more concerning. The company average is 2.89 out of 5, and more employees are rated 1 or 2 (40.3%) than 4 or 5 (32.9%). Customer Support, the largest department at 54 people, averages 2.47.

Attrition does at least skew towards weaker performers: 23.4% of those rated 1 leave, against 12.5% of those rated 4 or 5. From a retention standpoint this is comparatively healthy — the company is not haemorrhaging its best people. But it also means attrition is doing work that active performance management should be doing.

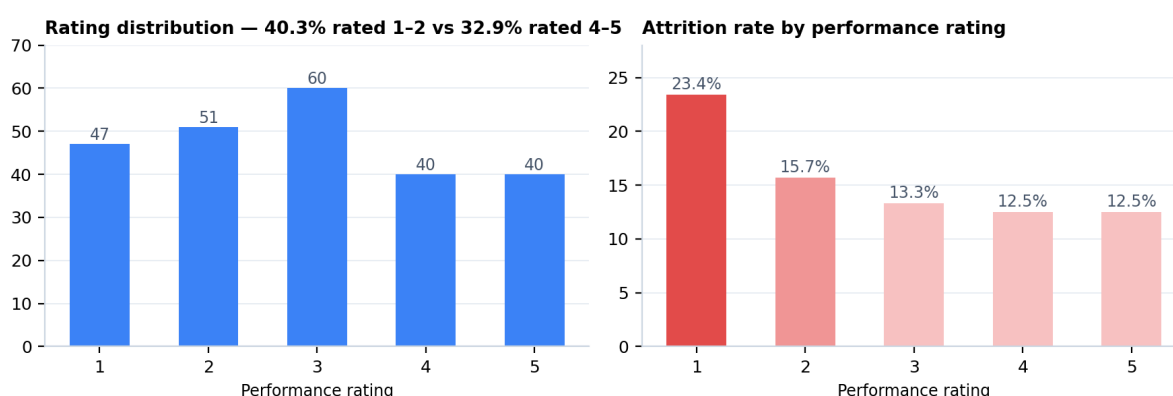


Figure 8 — Performance rating distribution and attrition rate by rating

Training does not appear to explain the variation. Training hours range narrowly from 35 to 44 across departments and show no relationship with rating: Operations receives the least training yet rates mid-range, while Customer Support receives above-average training and rates lowest. The issue is unlikely to be training volume.

Insight 6 — Pay is equitable; retention is not

The gender pay gap is 0.9% — KES 151,535 for men against KES 150,131 for women — which is negligible and reflects well on TalentCore's pay practices. Women also make up 51.9% of the workforce.

Retention tells a different story. Women leave at 19.8%, men at 11.1%: a gap of 8.7 percentage points. TalentCore pays equitably but does not retain equitably, and the pay data offers no explanation for the difference.

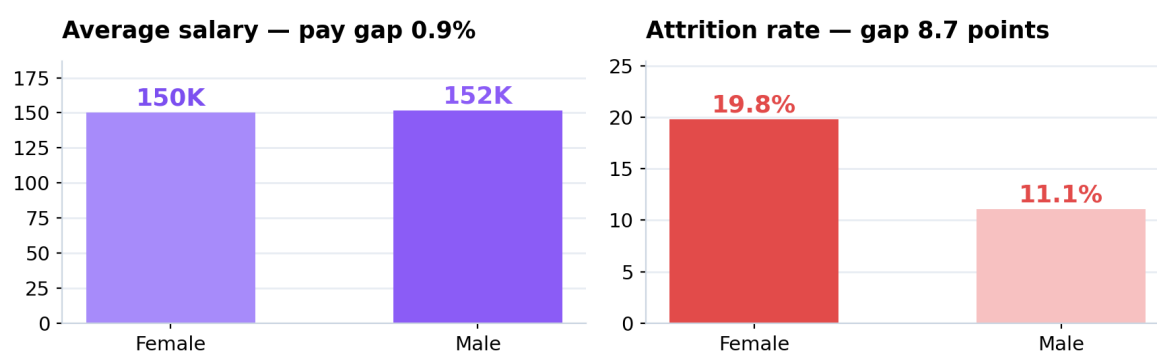


Figure 9 — Gender comparison: pay versus retention

Insight 7 — Qualifications do not track with pay

Master's degree holders are the lowest-paid group in the company at KES 139,519, below diploma holders at KES 158,135. They are also the most likely to stay, with attrition of 9.5%. Professional certification holders are the second-highest paid and the most likely to leave, at 23.5%.

This inverts the expected relationship and suggests that pay at TalentCore is set by role and department rather than by qualification — consistent with the finding that Engineering pays twice what Customer Support pays regardless of who occupies the role.

Master's holders earn least but stay longest

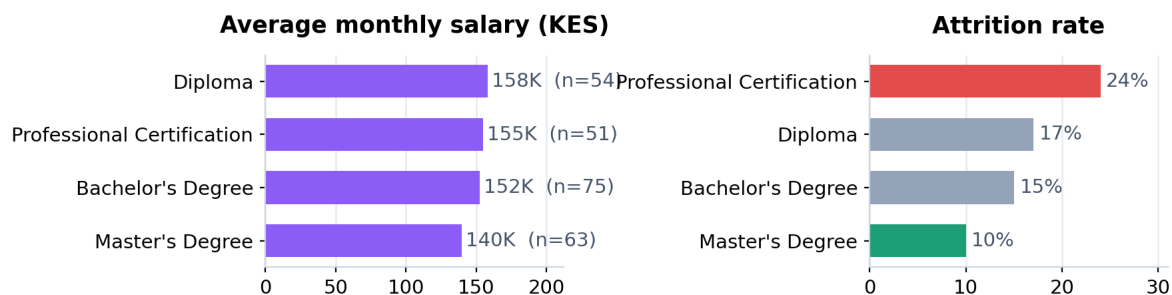


Figure 10 — Average salary and attrition rate by education level

9. Recommendations

Six recommendations follow from the analysis, ordered by expected impact. Each specifies an owner, a target and a metric so that progress can be measured against the dashboard rather than asserted.

R1 — Commission a structured exit review for 2024 departures

Eleven people left in 2024, concentrated in Finance and Customer Support, and the company has no recorded reason for any of them. At a conservative six months' salary per replacement, the 38 total departures represent approximately KES 34.4 million in cost. Interview the 2024 leavers where contactable and run stay interviews with current Finance and Customer Support staff.

Owner	Timeline	Target	Metric
CHRO / People Ops	Q1	Both departments below 15% attrition within two quarters	Attrition Rate by Department, dashboard page 3

R2 — Treat low overtime as an early-warning signal

The finding that non-overtime employees leave three times as often reframes a metric usually read as a wellbeing indicator. Build a watchlist of employees who work no overtime and hold a performance rating below 3, and require a manager conversation within 30 days of flagging. This should not become a mandate to work longer hours; the objective is to detect disengagement early, not to reward presenteeism.

Owner	Timeline	Target	Metric
Department heads	Ongoing	Flagged cohort retained at or above the company baseline over 12 months	Custom cohort filter on page 3

R3 — Audit the Customer Support operating model

Customer Support is the largest department at 54 people, the lowest paid at KES 113,909, the lowest rated at 2.47, and second-highest for attrition at 20.4%. Training hours are already at the company average, so additional training is unlikely to resolve it. The review should cover role design, span of control, career progression and pay banding rather than training volume.

Owner	Timeline	Target	Metric
COO / CHRO	Two quarters	Average rating of 2.8 and attrition below 15% within 12 months	Performance matrix, dashboard page 2

R4 — Introduce an 18-month retention checkpoint

With 27.7% attrition in the one-to-three-year band against 8.6% beyond five years, the company is losing people at the point of maximum sunk investment and minimum return. Introduce a structured checkpoint at 18 months covering career pathway, pay review and internal mobility, before the risk window opens.

Owner	Timeline	Target	Metric
People Ops	Q2	1-3 year band attrition below 18% within 12 months	Attrition by Tenure Band, dashboard page 3

R5 — Identify and transfer Nairobi's retention practices

Nairobi retains at 5.6% while Kisumu loses 22.9%, on near-identical headcount and pay. This is the largest unexplained variance in the dataset and the cheapest opportunity available: the practice that works already exists inside the company. Document Nairobi's management approach, onboarding and team structure, then pilot it in Kisumu.

Owner	Timeline	Target	Metric
Regional HR	Q2-Q3	Kisumu attrition below 15% within 12 months	Attrition by Office Location, dashboard page 3

R6 — Investigate the female attrition gap

Women leave at 19.8% against 11.1% for men, while the pay gap is only 0.9%. Because compensation has been ruled out as the driver, the investigation should examine progression, management practice, flexibility and role allocation. This requires data the company does not currently hold, which is itself the first finding.

Owner	Timeline	Target	Metric
CHRO / D&I lead	Q2	Gap narrowed to below 4 points within 12 months	Attrition Rate by Gender, dashboard page 4

R7 — Fix data capture at source

Nine of the twelve flagged records are exit dates that precede the hire date, and three leavers have no exit date at all. A single validation rule in the HR system — requiring a valid date format and rejecting any exit date earlier than the hire date — would eliminate three-quarters of the data quality issues found here and make future analysis materially more reliable.

Owner	Timeline	Target	Metric
People Ops / IT	Q1	Data completeness above 99% at the next refresh	Data Completeness %, dashboard page 5

10. Conclusion

TalentCore is a growing company with equitable pay, low historical attrition and a workforce that is, on balance, stable. The 2024 spike is real and warrants investigation, but it followed three years of exceptional retention and should be read as a signal to act rather than as evidence of crisis.

Two findings deserve particular attention because both invert the expected relationship. Employees who work no overtime leave three times as often as those who do, which suggests that engagement rather than workload drives departures. And the company's weakest performance scores sit in its largest and lowest-paid department, where more training has already been tried without effect.

The analysis was constrained by data quality. Three-quarters of the source file was padding, twelve of thirty-eight leavers cannot be placed on a timeline, and no exit reasons were recorded at all. Fixing capture at source is the least glamorous recommendation in this report and probably the most valuable, because it determines how much the next analysis will be able to say.

Appendix A — Detailed Figures

A.1 By department

Department	Headcount	Attrition	Avg salary (KES)	Avg rating	Avg training
Finance	37	21.6%	175,659	2.92	40
Customer Support	54	20.4%	113,909	2.47	38
Engineering	41	17.1%	219,861	3.00	40
Human Resources	39	12.8%	117,531	3.08	40
Sales	35	11.4%	151,600	3.11	44
Operations	37	8.1%	137,611	2.97	35
Company	243	15.6%	150,807	2.89	39

A.2 By office location

Office	Headcount	Attrition	Avg salary (KES)
Kisumu	48	22.9%	152,655
Mombasa	40	17.5%	153,088
Kampala	38	15.8%	152,795
Kigali	40	15.0%	150,123
Dar es Salaam	41	14.6%	149,600
Nairobi	36	5.6%	145,847

A.3 By year

Year	Hires	Exits	Headcount at 31 Dec	Annual attrition
2018	40	0	40	0.0%
2019	32	0	72	0.0%
2020	35	0	107	0.0%
2021	34	2	139	1.4%
2022	30	4	165	2.4%
2023	29	4	190	2.1%
2024	36	11	215	5.1%
2025 (partial)	7	5	217	2.3%

A.4 Attrition by segment

Segment	Group	n	Attrition
Overtime	No	110	24.5%
Overtime	Yes	133	8.3%

Segment	Group	n	Attrition
Tenure	1–3 years	65	27.7%
Tenure	3–5 years	60	13.3%
Tenure	5+ years	116	8.6%
Rating	1	47	23.4%
Rating	2	51	15.7%
Rating	3	60	13.3%
Rating	4	40	12.5%
Rating	5	40	12.5%
Gender	Female	126	19.8%
Gender	Male	117	11.1%
Education	Professional Certification	51	23.5%
Education	Diploma	54	16.7%
Education	Bachelor's Degree	75	14.7%
Education	Master's Degree	63	9.5%

Tenure band "under 1 year" contains only two employees and has been suppressed throughout the analysis as statistically meaningless.

A.5 Data quality summary

Metric	Value
Raw rows in source file	1,000
Padding rows removed	753
Duplicate Employee IDs removed	4
Records analysed	243
Records flagged with quality issues	12
Exit date before hire date	9
Marked as Left with no exit date	3
Active employees holding an exit date	0
Data completeness	95.1%